

B. Abraham's Great Escape

time limit per test: 1.5 seconds
 memory limit per test: 256 megabytes

Abraham is a brave explorer who goes where no other programmer has gone before. For his next expedition, he plans to investigate a peculiar maze. He knows that the maze is an $n \times n$ grid with an arrow in each cell that points in one of four directions: up, down, left and right. Abraham also knows that if he stands on an arrow, he will be forced to follow the arrows starting from that cell. Each arrow moves Abraham exactly 1 cell in the direction that it is pointing. If he reaches an arrow that points towards the outside of the maze, Abraham will escape the maze.

Abraham doesn't know how the arrows are arranged, so he wants to plan for multiple scenarios. He tasks you with finding an arrangement of arrows in the grid such that there are exactly k starting cells from which he can escape the maze.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 1000$). The description of the test cases follows.

The only line of each test case contains two integers n, k ($2 \leq n \leq 100, 0 \leq k \leq n^2$) — the size of the grid and the number of cells from which Abraham should be able to escape.

It is guaranteed that the sum of n^2 over all test cases does not exceed 10^5 .

Output

For each test case, do one of the following:

- If there exists a grid satisfying the requirement, print YES and then print n lines with n characters in each line indicating the direction of the arrows. Each character should be one of U (up), R (right), L (left), or D (down).
- Otherwise, declare that the task is impossible by printing NO.

If there are multiple solutions, print any of them.

You can output the answer in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

Example

input	Copy
3 2 4 3 5 2 3	
output	Copy
YES UU UU YES UUU RDR ULR NO	

Note

In the first test case, no matter which cell Abraham stands in initially, he will eventually exit the maze as he will move upward successively; thus, he can escape from all 4 cells, as required.

Codeforces Round 1056 (Div. 2)

Contest is running

01:21:41

Contestant



→ Submit?

Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	424
Problem B	848
Problem C	1484
Problem D	1696
Problem E	1908
Problem F	2544
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:38 from the first attempt



In the second test case, Abraham eventually escapes if he stands on one of the following:

- Any cell in the first row (all of which are U)
- Any cell in the third column (one of which is U and the other two R)

There is no other cell where he can stand and escape. We see that Abraham can escape from exactly 5 cells in this arrangement, as required.

In the third test case, it can be proved that there's no arrangement of arrows so that Abraham can escape from exactly 3 cells.

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